

Alfredo Jeong Hyun Park

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EXPERIENCE

Rubidex

Mar. 2023 – Present

Senior Fullstack Engineer

Condesa, Mexico City

- Led the architecture and development of **BMS Intel**, a large-scale Building Management System built with **Next.js**, **NestJS**, and **PostgreSQL**, enabling real-time industrial device control via **WebSockets**. bmsintel.com
- Led a **system-wide ID migration** (numeric UUID/string) across **100+ files**, enabling global scalability and cross-service consistency.
- Architected core backend systems including a **Work Order module** and a **command execution system** with UUID-based request tracking for reliable hardware control.
- Built shared platform infrastructure including a **centralized File Service** and a **hardware metadata engine** with GitHub synchronization and Redis caching.
- Implemented advanced authentication systems including **Paskeys (WebAuthn)**, backup codes, and Redis-based challenge flows.
- Architected and scaled a **multi-tenant admin platform** and production infrastructure using **Docker** and **CI/CD**, supporting multiple products including **Gridlock**. gridlock.co.

Listo.mx

Nov. 2021 – May 2022

Software Developer Intern

Roma, Mexico City

- Processed financial data to generate structured **PDF reports** and built dynamic UI components using **Django**.
- Implemented **automated tests** to improve reliability of production deployments.

PROJECTS

CityCommute

2024 – 2025

Lead Developer

Simulation & Visualization Engine

- Built a real-time urban mobility simulation modeling **thousands of concurrent agents** using **Python (Mesa)** and **FastAPI**.
- Implemented **A* pathfinding** and developed a real-time visualization system using **React** and **WebSockets**.
- Refactored and optimized **IBM Project-Multiagentes**, improving system performance and adding traffic analytics.

Ticket Crush

2025

Freelance

E-commerce & CMS

- Developed dynamic content systems using **Strapi** and designed reusable data models for event-driven platforms.

EDUCATION

Tecnológico de Monterrey

Aug. 2019 – Jul. 2024

B.S. in Computer Science and Technology

Mexico City

- **Coursework:** Data Structures, Algorithms, Advanced Algorithms, Multi-Agent Systems, Artificial Intelligence

SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C++, C#

Frameworks: React, Next.js, FastAPI, NestJS

Infrastructure: Docker, Redis, CI/CD, Linux

Tools: Git, Vite, Figma, Strapi